CRYSTAL
PRACTICE MANAGEMENT

CRYSTAL PM

HOW-TO GUIDE

Let Crystal Meet rooms join Google Meet

Give each room its own Google Workspace account and sign it in once, so Meet treats the room like a colleague instead of an unknown guest.

Google Meet refuses guests that arrive through an automated browser, which is how a Crystal Meet room joins. A room that is signed in as a Workspace user of its own is accepted, joins your company's meetings without knocking, and asks to join everyone else's under its own name. The Workspace half takes an admin about fifteen minutes for all rooms; the sign-in on each room's Pi takes five minutes over VNC from your laptop; nothing is plugged into the Pi.

STEP 1
A user per room
room1@, room2@, room3@, named after the rooms.

STEP 2
Their own unit
Sessions that never expire, no 2-step prompt.

STEP 3
Sign the room in
Once per Pi, on its screen.

STEP 4
Test it
An internal meeting, then an outside one.

BEFORE YOU BEGIN

Access. A Google Workspace super admin, or an admin who can create users and change security settings for an organizational unit; SSH to each room's Pi and a VNC viewer on your laptop for the one-time sign-in.

Cost. One Workspace seat per room. A single shared rooms account on all Pis is cheaper, but then every room appears under one name in calls and nobody can tell which room is which; this guide assumes one account per room.

Secrets. Each room user's password is typed once on its Pi and then kept with the room's other secrets. The signed-in session lives on the Pi in `/var/lib/croom/meet-profile`, readable by the room service only.

1 Create a Workspace user for each room

- **Add the user.** In the Admin console open **Directory › Users › Add new user**. First name `Room`, last name `1`, primary email `room1@crystalpm.com`. Set a strong password yourself rather than letting Google generate one you have to change on first sign-in.
- **Repeat** for `room2@crystalpm.com` and `room3@crystalpm.com`. The display name is what hosts see when the room joins, so keep it the room's name.
- **Nothing else is needed.** No Gmail, Drive or Calendar access is required for joining; any Workspace edition's seat works.

You should now see three users named Room 1, Room 2 and Room 3 in the user list.

2 Put them in a unit whose sessions never expire

- **Make the unit.** **Directory › Organizational units › Create organizational unit**, named `Meeting rooms`. Then select the three room users in the user list and use **Change organizational unit** to move them into it.
- **Keep them signed in.** **Security › Access and data control › Google session control**, select the Meeting rooms unit on the left, and set the web session duration to never expire (or the longest choice offered). Without this Google signs the rooms out on its schedule and someone has to sign them in again.
- **No second factor on the Pi.** **Security › Authentication › 2-Step Verification**, the Meeting rooms unit: allow users to turn it on, but do not enforce it, so the one-time sign-in needs only the password.
- **Optional.** Under **Apps › Google Workspace** you can turn Gmail and Drive off for the unit; Meet and Calendar stay on.

You should now see the three users inside Meeting rooms, with session control and 2-Step Verification set for that unit.

3 Sign each room in, once

- **Check the config.** The room configs from `deploy/rooms/` already carry `google_profile_dir: /var/lib/croom/meet-profile` under `meeting`. A room installed before this guide needs that line added to `/etc/croom/config.yaml`.
- **See the Pi's screen from your laptop.** Over SSH turn on VNC once with `sudo raspi-config nonint do_vnc 0`, then open a VNC viewer on your laptop (RealVNC Viewer or TigerVNC) to `pimeet-3.local` or the Pi's address and sign in with the Pi's username. The TV's screen appears in the window, so the Pi needs no keyboard or mouse of its own.
- **Stop the room service** so the browser is free: over SSH, `sudo systemctl stop croom`.
- **Open the sign-in.** Type `sudo -u pi DISPLAY=:0 /opt/croom/venv/bin/croom --sign-in-meet -c /etc/croom/config.yaml`, where pi is the user the service runs as, the one that ran the installer; the installer's completion text prints the exact command. Not as root: the profile must belong to the service's user. The room's own browser opens on the TV at Google's sign-in page.
- **Sign in in the VNC window** as that room's user, for example `room1@crystalpm.com`, accept Google's first-run prompts, and when the account page shows, close the browser window. The command prints "Profile saved".
- **Start the service again:** `sudo systemctl start croom`. Turn VNC off again if you prefer: `sudo raspi-config nonint do_vnc 1`.

You should now see "Profile saved" in the terminal, and the room service running again.

> **GOOD TO KNOW**
>
> The profile folder belongs to the room service's user and nobody else. Running the sign-in again later only re-uses it; if the session ever lapses, the room page says the sign-in has expired or was never done, and these same three commands fix it.

4 Test it

- **An internal meeting.** Start a meeting from a crystalpm.com account, paste its link into the room page's **Join with a link**, or book it on the room's calendar and press **Join now**. The room is in the call within about fifteen seconds; Meet shows it a **Join now** button, so nobody has to admit it.
- **An outside meeting.** Join a meeting hosted by a personal Google account or another company. The room asks to join under its own name, the room page says it is waiting, and the host admits it with **Ask to join**'s prompt.
- **Camera and microphone.** Both start in the state the room config asks for (**camera_default_on**, **mic_default_on**), whatever Meet remembered last time. Mute and camera buttons on the table page still work in the call.
- **From a terminal.** `sudo -u pi DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <meeting link> -c /etc/croom/config.yaml` opens the link from the signed-in profile and prints what Meet showed, with a screenshot under `/tmp`. It also prints a **Devices the browser sees** line with the microphones, speakers and cameras the browser can use. Stop the service first, as for the sign-in.

You should now see the room join the internal meeting on its own and the outside meeting after being admitted.

IN THE CALL

From the table a Meet call offers **Mute**, **Turn camera off**, **Raise hand** and **Captions**, and **Leave**: the room presses Meet's own keyboard shortcuts and reads each button's label back to see what happened. Meet has no chat, roster, reactions, view, pin, self view or host controls here, so the table draws no **Chat**, **Participants**, **Reactions** or **Host** button in a Meet call. A control Meet did not take answers in words on the screen, for example `Meet did not change the camera; try again`, and so does **Captions** while something on the TV that takes typing has the keyboard: `Meet's chat box has the keyboard; close it on the TV first`. The buttons are found by the English labels Meet showed at the time of writing, `"Turn off microphone"`, `"Turn on camera"`, `"Raise hand"` and `"Turn on captions"`, so a Meet whose labels differ hides the control instead of pressing the wrong thing. For the first real call open `http://<device>:8080/api/meeting/diagnostics` during it: for Meet it lists every label the page draws, what the room made of them and what has the keyboard; then check that **Mute** and **Turn camera off** each change in Meet with one tap on the table. The table waits two seconds for a button's label to change; a real camera may take longer to start, and then the table says Meet did not change the camera while the camera comes on a moment later (the room reads Meet again every five seconds), so note how long yours takes.

IF SOMETHING IS OFF

Troubleshooting

The room page says the sign-in has expired or was never done

Google's sign-in page came up instead of the meeting. Repeat step 3. If it keeps happening every couple of weeks, the session control in step 2 is not applied to the Meeting rooms unit.

The room page quotes "You can't join this video call"

The room is joining as a guest: either `google_profile_dir` is missing from `/etc/croom/config.yaml`, or the sign-in in step 3 was closed before it finished. Check the config, then repeat step 3.

The wrong name shows in the call

Meet shows the Google account's name. Change the user's first and last name in the Admin console, or sign the Pi in as the right room user.

Camera or microphone off in the call

The room sets both from the config before joining; if a device is missing, Meet's own default applies and the log says which button was not found. Check the camera and speakerphone are plugged into the Pi and seen by `arecord -l` and `v4l2-ctl --list-devices`.

The sign-in command says the service is using the profile

Stop the room service first with `sudo systemctl stop croom`, run the command again, and start the service when done.

Google asks for a phone or a second factor on the Pi

2-Step Verification is enforced for the room users. Set the Meeting rooms unit to allow but not enforce it (step 2), wait a few minutes, and try again.

GOOD TO KNOW

The room's browser is never disguised: it announces itself as an automated browser, which is exactly why a signed-in account is needed. To take a room's access away, suspend its user in the Admin console.